Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_22

1. Summary of Key Deviations

The analysis of Run_22 reveals a major, system-wide loss of flow, with a particularly unusual and severe signature in the bypass loop.

- **Flow_Bypass (Major):** The most significant anomaly is a large **negative offset** on the bypass flow, which reads as low as **-65.1%** compared to healthy. This indicates a massive reduction in the flow being actively managed by the bypass valve.
- **Flow_Pump (Moderate):** The pump flow shows a corresponding, significant **negative offset** of approximately **-32.8%**. This indicates a system-wide loss of flow originating at or before the pump.
- **System-Wide Pressure & Flow Loss (Minor to Moderate):** All other pressure and flow sensors (`p_Pump`, `p_FMU`, `p_Combustion`, `p_Shut`, `Flow_Engine1-4`, `Flow_PRV`) also show consistent minor to moderate **negative offsets**.
- **Engine Flows:** The engine flows are only slightly negative, suggesting the control system is prioritizing keeping them running despite the major loss of total flow.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Bypass Valve**	**Stuck Significantly Open (Internal Leak)**	This is the most compelling hypothesis. If the bypass valve is stuck open, it creates a permanent internal leak, diverting a large portion of the pump's output directly back to the inlet *before* it can be properly utilized or measured by the main flow sensors. This perfectly explains why both **Flow_Pump** (-32.8%) and especially **Flow_Bypass** (-65.1%) show such large negative deviations. The system is producing flow, but it's being immediately lost in a loop, starving the rest of the system.

2	**Main Fuel Pump**	**Major Internal Leakage / Degradation**	A degraded pump would explain the `Flow_Pump` drop of -32.8% and the general system-wide pressure loss. However, it doesn't as elegantly explain why the `Flow_Bypass` reading is so disproportionately low (-65.1%). A stuck-open valve provides a better explanation for the specific relationship between these two signals.
3	**Boost-Pressurised Tank / Inlet Line**	**Major Blockage**	A major blockage before the pump would starve it of fuel, causing a drop in `Flow_Pump` and `Flow_Bypass`. This is a plausible alternative for the physical fault, but again, doesn't explain the particularly severe drop in the bypass reading as well as a stuck valve does.
4	**Flow_Bypass & Flow_Pump Sensors**	**Dual Sensor Failure (Negative Offset)**	It is possible that both sensors have failed with a large negative offset. However, the fact that all other system pressures and flows are *also* showing a consistent negative trend makes a real, physical fault far more likely than a coincidental failure of multiple sensors.

5	**Catastrophic System-wide Leak**	**Major Rupture in Main Fuel Line**	A massive rupture in the main fuel line after the pump would cause a loss of pressure and flow. However, the unique and extreme negative deviation of the `Flow_Bypass` signal makes a fault localized to the bypass valve itself a more specific and fitting diagnosis.

3. Most Probable Hypothesis

The most probable root cause is that the **Bypass Valve is Stuck Significantly Open**, creating a major internal leak.

Reasoning: This single mechanical failure provides the most complete explanation for the unique data signature. When the bypass valve sticks open, a large portion of the pump's output is immediately "leaked" back to the pump inlet. This has two key effects seen in the data: 1) The overall flow available to the rest of the system drops, causing the `Flow_Pump` reading to be low (-32.8%). 2) The flow through the *measured and controlled* part of the bypass circuit, where the `Flow_Bypass` sensor is, also plummets dramatically (-65.1%) because most of the diverted flow is going through the unintended leak path. This one fault elegantly explains both major negative flow deviations, which other hypotheses struggle to reconcile.